INCH-POUND

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MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M16 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M16 Cartridge Actuated Thruster Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109 Quality Assurance Terms And Definitions
MIL-STD-413 Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head Division (CAGE Codes 19200 and 19203)

10521322	Thruster, Cartridge Actuated, M16 Metal Parts Assembly
8845243	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M16
8845244	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M16
9217557	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M16

U.S. Air Force

TPO 00-849-0214	Thruster, Cartridge Actuated, M16

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.3 and 6.3).

3.2 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to

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manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure. The cylinder and head shall withstand a hydrostatic pressure of 10,000 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.4.1).

3.5 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing and the following.

3.6.1 Fluid leakage. The thruster subassembly (less head, cartridge retainer, and spring) shall show no evidence of fluid leakage after a minimum of 16 hours, when kept in a vertical position with the open end of the cylinder pointing down.

3.6.2 Head and cylinder. The head and cylinder shall be assembled with a sufficient torque to lightly bottom components.

3.6.3 Presence of piston, shear pin. Upon completion of assembly, a load of 20 ± 2 pounds shall be applied to the locking pin to determine if the shear pin is properly engaged.

3.7 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.8 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.7.

3.9 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.9.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

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3.9.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.9.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77°F)
b. Altitude: sea level

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c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Pin, Firing, Drawing 8596416, covering a detail of Drawing 10521322.

Critical
None defined

Major

101	Concentricity of groove with body		Gage
102	Concentricity of striker with body		Gage
103	True position of pin hole	Gage	
104	Diameter of body, min	Gage	
105	Diameter of groove	Gage	

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106	Diameter of striker	Gage	
107	Diameter of pin hole, max	Gage	
108	Length of striker, min	Gage	
109	Profile of striker	Gage	
110	Profile of groove	Gage	
111	Metal defective	Visual	
112	Protective finish inadequate or defective		Visual
113	Surface finish of body improper		Visual
114	Surface finish of striker improper		Visual
115	Foreign matter	Visual	
116	Burr or sharp corners	Visual	

Minor

201	Diameter of body, max	Gage	
202	Diameter of pin hole, min	Gage	
203	Total length	Gage	
204	Length of striker, max	Gage	
205	Location of pin hole	Gage	
206	Depth of slot in base	Gage	
207	Width of slot in base	Gage	
208	Width of groove	Gage	
209	Location of groove	Gage	
210	Location of slot in base	Gage	
211	Marking missing, incorrect, or illegible		Visual
212	Radii incorrect	Gage	
213	Surface finish improper, except as otherwise classified		Visual

4.5.2.2 Piston, Floating, Drawing 8596417, covering a detail of Drawing 10521322.

Critical
None defined

Major

101	Concentricity of groove with body		Gage
102	Pitch diameter of thread, max		Gage
103	Major diameter of thread, max		Gage
104	Diameter of groove	Gage	
105	Diameter of body, min	Gage	
106	Length of effective thread, min		Gage
107	Profile of groove	Gage	
108	Metal defective	Visual	
109	Protective finish inadequate or defective		Visual
110	Thickness	Gage	
111	Burr or sharp corner	Visual	
112	Foreign matter	Visual	

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Minor

201	Pitch diameter of thread, min	Gage
202	Diameter of body, max	Gage
203	Diameter of hole through body	Gage
204	Diameter of wrench holes	Gage
205	Depth of wrench holes	Gage
206	Radii or chamfer incorrect	Gage
207	Width of groove	Gage
208	Depth of threaded cavity	Gage
209	Location of wrench holes	Gage
210	Location of groove	Gage
211	Marking missing, incorrect, or illegible	Visual
212	Surface finish improper	Visual

4.5.2.3 Piston, Drawing 8596418, covering a detail of Drawing 10521322.

Critical

None defined

Major

101	Concentricity of pitch diameter of thread with body	Gage
102	Concentricity of head with body	Gage
103	Concentricity of groove with head	Gage
104	Pitch diameter of thread, max	Gage
105	Minor diameter of thread, max	Gage
106	Diameter of head, min	Gage
107	Diameter of body, min	Gage
108	Diameter of groove	Gage
109	Length of effective thread, min	Gage
110	Width across slot, min	Gage
111	Width of slot	Gage
112	Profile of groove	Gage
113	Metal defective	Visual
114	Protective finish inadequate or defective	Visual
115	Surface finish of body improper	Visual
116	Burr or sharp corner	Visual
117	Foreign matter	Visual

Minor

201	Pitch diameter of thread, min	Gage
202	Diameter of undercut, min, if applicable	Gage
203	Diameter of head, max	Gage
204	Diameter of body, max	Gage
205	Diameter of hole through head	Gage
206	Depth of threaded cavity, max	Gage
207	Total length	Gage

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208	Length of head	Gage	
209	Width of groove	Gage	
210	Width across slot, max	Gage	
211	Location of slot	Gage	
212	Location of hole through head		Gage
213	Width of undercut, max, if applicable		Gage
214	Location of groove	Gage	
215	Marking missing, incorrect, or illegible		Visual
216	Radius or chamfer incorrect		Gage
217	Surface finish improper, except as otherwise classified		Visual

4.5.2.4 Pin, Shear, Drawing 8596419, covering a detail of Drawing 10521322.

Critical
None defined

Major

101	Diameter, max	Gage	
102	Total length, min	Gage	
103	Metal defective	Visual	
104	Protective finish inadequate or defective		Visual
105	Burr	Visual	
106	Foreign matter	Visual	

Minor

201	Total length, max	Gage
202	Surface finish improper	Visual

4.5.2.5 Clevis, Drawing 8596421, covering a detail of Drawing 10521322.

Critical
None defined

Major

101	True position of hole	Gage	
102	Pitch diameter of thread, min		Gage
103	Major diameter of thread, min		Gage
104	Diameter of hole, max	Gage	
105	Diameter of undercut, min	Gage	
106	Metal defective	Visual	
107	Protective finish inadequate or defective		Visual
108	Burr or sharp corner	Visual	
109	Foreign matter	Visual	

Minor

201	Pitch diameter of thread, max	Gage

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202	Diameter of hole, min	Gage	
203	Depth of slot	Gage	
204	Width of clevis	Gage	
205	Width of slot	Gage	
206	Location of undercut	Gage	
207	Location of slot	Gage	
208	Width of undercut	Gage	
209	Location of hole	Gage	
210	Height of clevis	Gage	
211	Radius or chamfer incorrect		Gage
212	Marking missing, incorrect, or illegible		Visual
213	Surface finish improper	Visual	

4.5.2.6 Pin, Locking, Drawing 8596422, covering a detail of Drawing 10521322.

Critical
None defined

Major

101	Diameter of hole, max	Gage	
102	Location of hole	Gage	
103	Width, min	Gage	
104	Thickness, min	Gage	
105	Metal defective	Visual	
106	Protective finish inadequate or defective		Visual
107	Burr	Visual	
108	Foreign matter	Visual	

Minor

201	Diameter of hole, min	Gage	
202	Total length	Gage	
203	Width, max	Gage	
204	Thickness, max	Gage	
205	Marking missing, incorrect, or illegible		Visual
206	Surface finish improper	Visual	

4.5.2.7 Pin, Shear, Drawing 8596423, covering a detail of Drawing 10521322.

Critical
None defined

Major

101	Diameter, max	Gage	
102	Total length, min	Gage	
103	Metal defective	Visual	
104	Protective finish inadequate or defective		Visual

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105	Burr	Visual
106	Foreign matter	Visual

Minor

201	Total length, max	Gage
202	Surface finish improper	Visual

4.5.2.8 Spring, Drawing 8596424, covering a detail of Drawing 10521322.

Critical
None defined

Major

101	Load at short compressed length	Gage
102	Load at long compressed length	Gage
103	Solid length, max	Gage
104	Ends incorrect	Visual
105	Protective finish inadequate or defective	Visual
106	Burr, sharp edges	Visual
107	Foreign matter	Visual

Minor

201	Outside diameter	Gage
202	Fit free into hole	Gage
203	Fit free over rod	Gage

4.5.2.9 Head, Drawing 10521324, covering a detail of Drawing 10521322.

Critical

1	Surface finish of long cavity improper	Gage

Major

101	Pitch diameter of large thread, max	Gage
102	Minor diameter of large thread, max	Gage
103	Pitch diameter of thread inlet, max	Gage
104	Minor diameter of thread inlet, max	Gage
105	Concentricity of large cavity with pitch diameter of large thread	Gage
106	Concentricity of boss with pitch diameter of large thread	Gage
107	Concentricity of hole through boss with long cavity	Gage
108	Concentricity of long cavity with pitch diameter of large thread	Gage
109	Concentricity of lugs	Gage
110	Perpendicularity of boss face with pitch diameter of large thread	Gage
111	Symmetry of lug flats with pitch diameter of large thread	Gage

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112	True position of lugs	Gage	
113	True position of pin hole	Gage	
114	Diameter of undercut, large thread, max		Gage
115	Diameter of large cavity, max		Gage
116	Diameter of hole through boss, min		Gage
117	Diameter of long cavity, max		Gage
118	Diameter of lugs, min	Gage	
119	Depth of counterbore, small end		Gage
120	Depth to bottom of large cavity, min		Gage
121	Thickness through boss	Gage	
122	Height of boss	Gage	
123	Length of effective thread, small end, min		Gage
124	Length of thread, small end, max		Gage
125	Diameter of chamfer, small end		Gage
126	Angle of chamfer, small end	Gage	
127	Internal radii	Gage	
128	Thread damaged or not full	Visual	
129	Metal defective	Visual	
130	Protective finish inadequate or defective		Visual
131	Surface finish of chamfers, small end, and boss incorrect		Visual
132	Burr or sharp corners	Visual	
133	Foreign matter	Visual	

Minor

201	Pitch diameter of large thread, min		Gage
202	Pitch diameter of small thread, min		Gage
203	Pitch diameter of setscrew thread		Gage
204	Minor diameter of setscrew thread, max		Gage
205	Diameter of large cavity, min		Gage
206	Diameter of hole through boss, max		Gage
207	Diameter of long cavity, min		Gage
208	Diameter of lugs, max	Gage	
209	Diameter of counterbore, small end		Gage
210	Diameter of pin hole	Gage	
211	Diameter of small end, external		Gage
212	Diameter of large end, external		Gage
213	Diameter of boss	Gage	
214	Diameter of inlet cavity	Gage	
215	Depth of bottom of large cavity, max		Gage
216	Depth to bottom of thread undercut		Gage
217	Depth of effective thread, setscrew, min		Gage
218	Total length	Gage	
219	Length of trunnion	Gage	
220	Length of lugs	Gage	
221	Width of trunnion flats	Gage	

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222	Width across trunnion flats		Gage
223	Location of pin hole	Gage	
224	Location of lugs	Gage	
225	Length of large body	Gage	
226	Width of thread undercut	Gage	
227	Depth of setscrew hole, max	Gage	
228	Marking missing, incorrect, or illegible		Visual
229	Radius or chamfer incorrect, except as otherwise classified		Gage
230	Surface finish improper, except as otherwise classified		Visual

4.5.2.10 Retainer, Cartridge, Drawing 10521325, covering a detail of Drawing 10521322.

Critical
None defined

Major

101	Concentricity of cavity with bore		Gage
102	Concentricity of body with bore		Gage
103	Diameter of bore, max	Gage	
104	Diameter of cavity, max	Gage	
105	Diameter of body, min	Gage	
106	Small outside diameter, min		Gage
107	Depth of cavity, max	Gage	
108	Internal radii incorrect	Gage	
109	Metal defective	Visual	
110	Protective finish inadequate or defective		Visual
111	Burr or sharp corner	Visual	
112	Foreign matter	Visual	

Minor

201	Diameter of bore, min	Gage	
202	Diameter of cavity, min	Gage	
203	Small outside diameter, max		Gage
204	Diameter of body, max	Gage	
205	Depth of cavity, min	Gage	
206	Total length	Gage	
207	Length of small outside diameter		Gage
208	Marking missing, incorrect, or illegible		Visual
209	External radii incorrect	Gage	
210	Surface finish improper	Visual	

4.5.2.11 Cylinder and Pellet Assembly, Drawing 10521323, covering a detail of Drawing 10521322.

Critical
None defined

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Major			
101	Concentricity of cavity with pitch diameter of thread		Gage
102	Concentricity of thread pilot with pitch diameter of thread		Gage
103	Concentricity of bore with cavity		Gage
104	Concentricity of groove with bore		Gage
105	Pitch diameter of large thread, min		Gage
106	Major diameter of large thread, min		Gage
107	Diameter of cavity, max	Gage	
108	Diameter of bore, max	Gage	
109	Diameter of internal undercut, max, if applicable		Gage
110	Diameter of thread pilot, min		Gage
111	Diameter of groove	Gage	
112	Depth of slot, max	Gage	
113	Length of effective thread, min, large		Gage
114	Length of thread, max, large		Gage
115	Width of slot, max	Gage	
116	Profile of groove	Gage	
117	Protrusion of pellets, max	Gage	
118	Pellet loose, missing, or improperly inserted		Visual/Manual
119	Thread damaged	Visual	
120	Metal defective	Visual	
121	Protective finish inadequate or defective		Visual
122	Surface finish of cavity and bore improper		Visual/Gage
123	Burr or sharp corners	Visual	
124	Foreign matter	Visual	
Minor			
201	Pitch diameter of large thread, max		Gage
202	Pitch diameter of setscrew thread		Gage
203	Minor diameter of setscrew thread, max		Gage
204	Diameter of cavity, min	Gage	
205	Diameter of bore, min	Gage	
206	Diameter of shear pin hole	Gage	
207	Diameter of thread pilot, max		Gage
208	Diameter of body	Gage	
209	Depth of setscrew hole, max	Gage	
210	Depth of slot, min	Gage	
211	Depth of effective thread setscrew, min		Gage
212	Length of thread pilot	Gage	
213	Total length	Gage	
214	Length of flat	Gage	
215	Length of bore	Gage	
216	Width of slot, min	Gage	
217	Width of groove	Gage	
218	Width across flats	Gage	

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219	Location of groove	Gage	
220	Location of setscrew hole	Gage	
221	Location of slot	Gage	
222	Location of flats	Gage	
223	Width of internal undercut, if applicable		Gage
224	Marking missing, incorrect, or illegible		Visual
225	Radii or chamfers incorrect		Gage
226	Surface finish improper, except as otherwise classified		Visual
227	Location of pellets	Gage	

4.5.2.12 Assembly, Drawing 10522631.

Critical

1	Protrusion of firing pin	Gage

Major

101	Any part missing, not otherwise classified		Visual
102	Improper assembly	Visual	
103	Material defective	Visual	
104	Foreign matter	Visual	

Minor

201	Protective finish inadequate or defective		Visual
202	Protective plug missing	Visual	
203	Marking missing, incorrect, or illegible		Visual

4.5.2.13 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8845244.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual	
202	Box damaged	Visual	
203	Marking missing, incorrect, or illegible		Visual
204	Cut, damaged, or improperly sealed bag		Visual

4.5.2.14 Box, Packing, Ammunition, Drawing 8845243.

Critical
None defined

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Major
None defined

Minor
201 Contents loose Visual/Manual
202 Strapping missing, loose, broken, or improperly assembled Visual/Manual
203 Contents of box exposed Visual
204 Box damaged Visual
205 Marking missing, incorrect, or illegible Visual
206 Number of units incorrect Visual

4.5.2.15 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217557.

Critical
None defined

Major
None defined

Minor
201 Contents loose Manual/Visual
202 Carton broken or damaged Visual
203 Number of units incorrect Visual
204 Marking missing, incorrect, or illegible Visual
205 Tape incomplete Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each cylinder and head shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Fluid leakage. Each subassembly shall be subjected to a fluid leakage test. If an appreciable amount of fluid accumulates after the minimum test period, the unit shall be rejected. If after the minimum test period there is only a thin film of fluid covering the piston face, then the piston face shall be wiped clean and the subassembly retested for another minimum period.

4.5.3.4 Presence of piston shear pin. Each assembly shall be subjected to a test to determine the presence of a shear pin in the piston lock. Failure of any unit to comply with the requirements of 3.6.3 shall be cause for rejection of the unit.

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4.5.3.5 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.5 shall be cause for rejection of the lot.

4.5.3.6 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.7 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.7. Failure of any unit in the inspection sample to comply with 3.8 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to application of protective finish the cylinder and head shall be mounted in a device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

4.6.3 Fluid leakage. Prior to assembly of the head, cartridge retainer, and spring, the subassembly shall be positioned vertically with the open end of the cylinder pointing down, for a minimum of 16 hours.

4.6.4 Presence of piston shear pin. Upon completion of assembly, the thruster shall be secured in a device and a load of 20 ± 2 pounds shall be applied to the locking pin to determine if the shear pin is properly engaged in the locking pin.

4.6.5 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6.6 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.7 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

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4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8845243 and 8845244. Level B shall be in accordance with Drawings 8845244 and 9217557.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-849-0214.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special instruction requirements (see 4.3)

f. Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E22)

Review activity:
Air Force - 70

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